

Winning Tickets Are Not Posterior Modes: Evidence from Posterior Support and Magnitude Controls

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Abstract

The lottery ticket hypothesis suggests that dense neural networks contain sparse subnetworks that can be trained in isolation when rewound to their original initialization. A tempting Bayesian interpretation is that these tickets correspond to posterior modes, or to sparse supports induced by posterior basins. We make this interpretation falsifiable: posterior-induced masks must match IMP tickets better than local magnitude, dense-trajectory, and pruning-process controls, not merely better than uniform random masks. Across five-seed MNIST and Fashion-MNIST sparsity sweeps, and a CIFAR-10 ResNet-20 epoch-1 rewind setting where IMP exceeds dense accuracy, SGLD, SGHMC, cyclical SGLD, SWAG, diagonal/KFAC Laplace, low-rank Hessian-plus-diagonal Laplace, exact last-layer Laplace, selected/tensor-block full-covariance Laplace, joint-group full-covariance Laplace, and subspace HMC all fail this stronger criterion. Posterior supports often beat random masks, but they remain tied to chain-start magnitude, dominated by dense magnitude, or closer to rewind controls than to IMP. Direct mode/ticket audits agree: posterior samples can match IMP tickets in logit or activation CKA while failing layer-sparsity and Hamming-distribution thresholds, and their basin count does not match ticket diversity. Mechanistic controls point instead to trajectory and pruning-process structure: IMP-only residual support recovers much of the fixed-mask training gap, while cross-seed transfer, posterior uncertainty, score-bin controls, overlap-matched process controls, and tensor-matched round-exclusion interventions do not rescue a posterior-mode account. In these settings, winning tickets are trajectory and pruning-process objects, not Bayesian posterior-mode supports. Exact dense full-network full-covariance CIFAR posterior baselines remain an open limitation.

1 Introduction

The lottery ticket hypothesis (LTH) states that a dense randomly initialized network contains sparse subnetworks that can be trained to match the dense network when reset to their original initialization [4]. This result raises a basic question about the object discovered by iterative magnitude pruning (IMP): is a winning ticket a combinatorial artifact of the optimization trajectory, or does it reflect a deeper probabilistic structure of the posterior over weights?

A Bayesian reading is attractive. SGD and its noisy variants have long been connected to approximate Bayesian inference [16, 11]. Bayesian deep learning further emphasizes that useful neural network solutions are distributed across broad posterior regions rather than isolated points [17, 7]. If winning tickets correspond to posterior modes or posterior-basin supports, LTH would be evidence for a selection view of learning: pruning would reveal posterior structure already present in the trained model.

This paper tests that interpretation directly. We ask:

Do posterior samples or posterior basins induce sparse supports that explain IMP winning-ticket masks beyond dense magnitude, initialization magnitude, rewind magnitude, chain-start magnitude, and pruning-process controls?

The paper contributes three pieces of evidence. First, it states the posterior-mode explanation as a support-equivalence claim with operational gates: posterior-induced supports must beat local magnitude controls, not just random masks, and mode/ticket distributions must match in both mask and function space. Second, it evaluates these gates across repeated MNIST/Fashion-MNIST sweeps and CIFAR-10 ResNet-20

posterior approximations, including stochastic samplers, Laplace variants, SWAG, subspace HMC, and direct mode/ticket distribution probes. Third, after the posterior account fails these gates, it tests a mechanistic alternative: IMP tickets are better explained by dense-trajectory and pruning-process structure, especially the IMP-only residual support.

The conclusion is negative but scoped. Posterior masks beat random masks, but they track the magnitude support of the dense checkpoint or the chain start. When chains move, they move away from the chain start without moving toward the IMP ticket. When function-space CKA is high, mask-distribution and basin-count criteria still fail. When process controls are added, the strongest positive signal is the IMP-selected residual, not a posterior-mode support.

2 Related Work

Lottery tickets and pruning. LTH was introduced by Frankle and Carbin [4]. Follow-up work connected LTH to rewinding and linear mode connectivity [5]. Random weighted networks with learned masks further emphasized the selection view of learning [13]. Single-shot pruning methods such as SNIP [9] and SynFlow [15] provide important non-Bayesian controls.

Mode connectivity and subspaces. Low-loss connectivity between neural network solutions challenges the view that trained models are isolated modes [6, 2]. For lottery tickets specifically, Paul et al. [12] argue that IMP masks encode axial subspaces that intersect matching linearly connected regions, which is a direct non-Bayesian competitor to a posterior-mode explanation. Permutation symmetries and re-basin methods further complicate parameter-space mode counting [3, 1]. Our experiments therefore include function-space clustering and linear barrier probes in addition to mask overlap.

Bayesian neural networks. SGLD provides a simple route from SGD to approximate posterior sampling [16]. SGD itself can be interpreted as approximate Bayesian inference in a stationary regime [11]. Practical posterior approximations such as SWAG [10] and exact small-model sampling studies [7] show that posterior geometry is structured and nontrivial. PAC-Bayesian and Bayesian LTH analyses [14, 8] motivate the need for controls: Bayesian language alone does not imply that IMP masks are posterior-mode supports.

Our contribution is therefore not a new posterior approximation or a claim that Bayesian inference is uninformative for pruning. It is a controlled support-equivalence test: if posterior modes explain winning tickets, their sparse supports should match IMP tickets after conditioning on dense magnitude, rewind magnitude, trajectory magnitude, and pruning-process alternatives.

3 Operational Test

Let m_{IMP} denote the mask found by IMP at a fixed sparsity. For a posterior sample $\theta^{(s)}$, the simplest posterior-induced mask is

$$m^{(s)} = \text{TopK}(|\theta^{(s)}|),$$

with the same number of kept weights as m_{IMP} . We also construct aggregate posterior masks from posterior mean, RMS, signal-to-noise ratio, high variance, and low variance.

We compare all masks by support Jaccard similarity to m_{IMP} :

$$J(m, m_{\text{IMP}}) = \frac{|m \cap m_{\text{IMP}}|}{|m \cup m_{\text{IMP}}|}.$$

The key controls are random same-sparsity masks, initialization magnitude masks, dense trained magnitude masks, rewind magnitude masks, chain-start magnitude masks for posterior chains, SNIP/SynFlow masks, fixed-mask retraining controls, channel-aligned direct mode/ticket controls, and process-residual interventions. The positive posterior-mode interpretation must satisfy a stronger condition than beating random: posterior masks must explain IMP beyond the chain-start, dense, rewind, trajectory, and process controls, and posterior samples must move support in a way that increases IMP alignment.

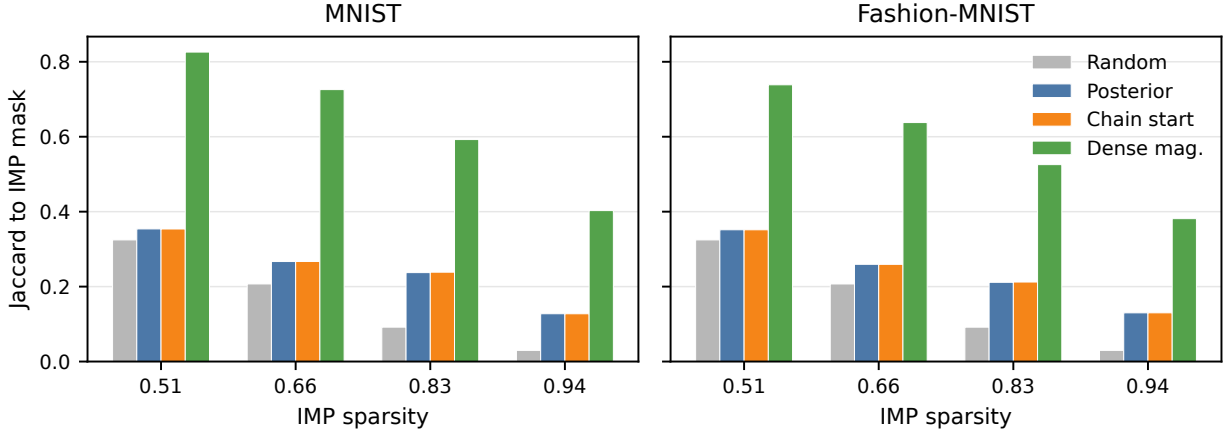


Figure 1: Gate1 control comparison on MNIST and Fashion-MNIST. Posterior masks beat random masks, but they track chain-start magnitude and remain far below dense magnitude across sparsities.

4 Current Results

4.1 Random Is Too Weak a Baseline

Tables 1 and 2 show the current Gate1 sparsity sweeps. Each row aggregates five seeds, three independent dense-start SGLD chains per seed, and posterior masks at the matched IMP sparsity.

Table 1: MNIST 5-seed Gate1 sparsity sweep. Posterior masks beat random but fail against stronger controls at every sparsity. “Post-Chain” is the Jaccard overlap between posterior masks and their chain-start magnitude masks.

Config	Runs	Sparsity	Posterior	Random	Chain Start	Post-Chain	Dense Mag	Gate1
r2 p0.30	5	0.5100	0.3539	0.3246	0.3537	0.9303	0.8259	Fail
r3 p0.30	5	0.6570	0.2669	0.2071	0.2670	0.9115	0.7258	Fail
r5 p0.30	5	0.8319	0.2373	0.0918	0.2385	0.9298	0.5924	Fail
r8 p0.30	5	0.9424	0.1275	0.0298	0.1275	0.9423	0.4028	Fail

Table 2: Fashion-MNIST 5-seed Gate1 sparsity sweep. The failure mode matches MNIST. At the highest sparsity, IMP accuracy drops below the dense model, so that row should be interpreted as a high-sparsity stress case.

Config	Runs	Sparsity	Posterior	Random	Chain Start	Post-Chain	Dense Mag	Gate1
r2 p0.30	5	0.5100	0.3516	0.3246	0.3516	0.9314	0.7385	Fail
r3 p0.30	5	0.6570	0.2595	0.2070	0.2594	0.9160	0.6378	Fail
r5 p0.30	5	0.8319	0.2114	0.0917	0.2122	0.9212	0.5255	Fail
r8 p0.30	5	0.9424	0.1300	0.0297	0.1300	0.9408	0.3813	Fail

Posterior masks beat random support at every tested MNIST/Fashion-MNIST sparsity, but the random gap is explained by chain-start magnitude. On MNIST and Fashion-MNIST, posterior-minus-chain-start gaps stay near zero while posterior-to-chain-start overlap remains around 0.91–0.94. Dense magnitude is much closer to IMP, with dense-minus-posterior gaps of 0.275–0.472 on MNIST and 0.251–0.387 on Fashion-MNIST. The same pattern holds in the generated mode-distribution audit: posterior supports beat uniform random masks in 58 of 59 grouped comparisons, but none of the 59 posterior-versus-chain-start comparisons

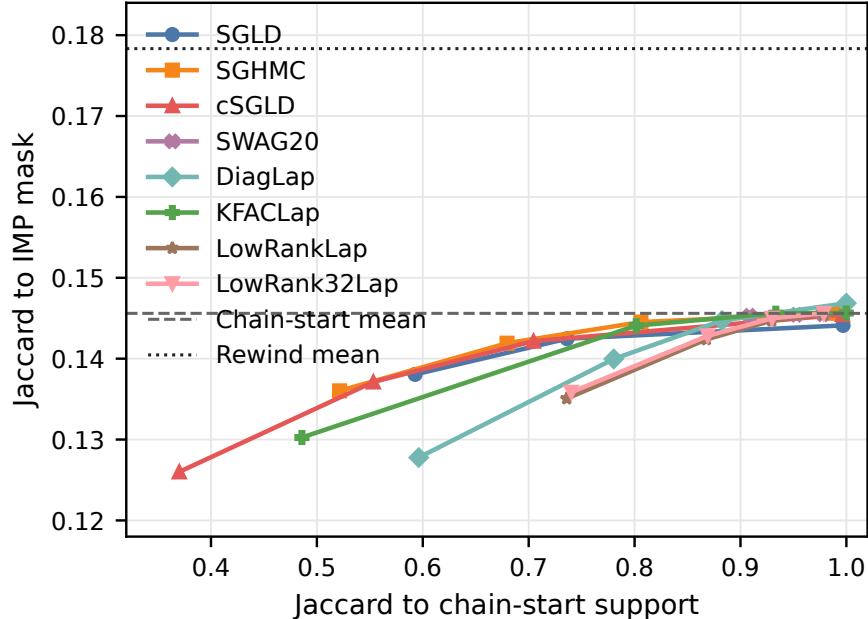


Figure 2: CIFAR-10 ResNet-20 epoch-1 rewind movement diagnostics. Across SGLD, SGHMC, cyclical SGLD, full-network SWAG, diagonal Laplace, KFAC-style Laplace, and low-rank Hessian-plus-diagonal Laplace, moving support away from the chain start does not increase overlap with the IMP mask.

beats the matched chain-start support by more than 0.005 Jaccard: 43 are practically tied to the control, 15 favor chain-start magnitude, and one is mixed. Rewind magnitude is closer to the IMP ticket than posterior support in 55 of 57 grouped comparisons.

4.2 Sampler Movement Does Not Point Toward Tickets

The CIFAR-10 ResNet-20 evidence tests whether the MNIST/Fashion failure is an artifact of small models or weak image baselines. In the long-budget epoch-1 rewind row, dense accuracy is about 88.5% and IMP reaches about 89.8% at 83.2% sparsity, yet SGLD, SGHMC, cyclical SGLD, SWAG, diagonal Laplace, KFAC-style Laplace, low-rank Hessian-plus-diagonal Laplace, and subspace HMC all fail the same control. Increasing sampler scale often lowers posterior-to-chain-start overlap, but posterior-to-IMP overlap does not improve and the epoch-1 rewind support remains closer to IMP.

The negative movement result is broad across posterior families. The sampler grid includes dense-start and independent-start multi-chain SGLD, SGHMC, cyclical SGLD, a 20-snapshot full-network SWAG movement row, diagonal and KFAC-style Laplace, rank-16, rank-32, rank-64, and rank-128 top-Hessian low-rank Laplace movement rows, exact final-head Laplace, selected and tensor-block-diagonal full-covariance Laplace, joint-group Laplace, and low-dimensional HMC in random, trajectory, and top-Hessian subspaces. Full-batch HMC and an exact dense full-network Laplace sanity check on small digits MLPs validate higher-fidelity small-model code paths while preserving the same qualitative support failure.

4.3 Richer Covariance Still Fails the Support Gate

The covariance objection is the strongest bounded limitation. We therefore push exact covariance as far as the workstation budget permits. We use a 22,064-parameter block-diagonal probe covering 22,064 trainable weights; at scale 10^{-4} sample accuracy remains 88.1%, global posterior-to-chain-start support falls to 0.8287, the selected-block posterior-to-IMP support is below the selected-block chain-start control by 0.0114 on average, the global posterior-minus-chain gain is only 0.0036, and rewind remains closer by 0.0292. Thus, this 22,064-parameter block-diagonal row moves support but not toward tickets.

Sampler	LR/Scale/Step	Dense	IMP	Post.	Chain	Post-chain	Rewind
SGLD	10^{-6}	0.8845	0.8993	0.1425	0.1441	0.7362	0.1784
SGLD	3×10^{-6}	0.8845	0.8993	0.1381	0.1441	0.5928	0.1784
cSGLD	10^{-6}	0.8852	0.8960	0.1422	0.1454	0.7046	0.1789
DiagLap	10^{-3}	0.8849	0.8980	0.1447	0.1469	0.8826	0.1787
DiagLap	10^{-2}	0.8849	0.8980	0.1278	0.1469	0.5961	0.1787
KFACLap	10^{-2}	0.8860	0.8956	0.1303	0.1456	0.4859	0.1775
LowRank32Lap	10^{-2}	0.8853	0.8964	0.1358	0.1457	0.7402	0.1795
LowRank64Lap	10^{-2}	0.8874	0.8954	0.1339	0.1433	0.7397	0.1766
LowRank128Lap	10^{-2}	0.8867	0.8967	0.1351	0.1453	0.7358	0.1780
SWAG20	64	0.8859	0.8973	0.1453	0.1455	0.9086	0.1782
RandSubHMC	3×10^{-3}	0.8866	0.8965	0.1440	0.1440	0.9766	0.1779
TrajSubHMC	10^{-3}	0.8849	0.8961	0.2290	0.2292	0.9915	0.1792
Hess16SubHMC	3×10^{-4}	0.8873	0.8974	0.1468	0.1468	0.9994	0.1799
Hess32SubHMC	3×10^{-4}	0.8868	0.8973	0.1461	0.1461	0.9993	0.1788

Table 3: Representative five-seed CIFAR-10 ResNet-20 epoch-1 rewind movement diagnostics at r5 p0.30. Posterior support either remains near the chain-start magnitude support or moves away from it, but posterior-to-IMP overlap (Post.) does not exceed the matching chain-start control. TrajSubHMC is higher because its chain-start control is the matched dense trajectory magnitude support; the full generated movement summary is in Appendix Table 4.

Wider exact covariance rows give the same answer, and this 68,144-parameter block-diagonal row, covering 68,144 trainable weights, reports that sample accuracy remains 88.0%, global support falls further to 0.7400, selected-block posterior-minus-chain remains negative at -0.0050, global posterior-minus-chain gain shrinks to 0.0010, and rewind remains closer by 0.0319, and this 68,144-parameter block-diagonal row remains negative. The 68,144-parameter joint-group row keeps the same 68,144 trainable weights while including cross-tensor covariance inside groups; sample accuracy remains 88.1%, global posterior-to-chain-start support falls to 0.7148, block posterior-minus-chain remains -0.0050, global posterior-minus-chain is 0.0015, and rewind remains closer by 0.0311, so this 68,144-parameter joint-group row also fails. The next row is this 86,576-parameter joint-group row, covering 86,576 trainable weights, and it preserves sample accuracy at 88.3%; global posterior-to-chain-start support is 0.7863, block posterior-minus-chain stays negative at -0.0023, global posterior-minus-chain stays 0.0006, and rewind remains closer by 0.0317. The largest row is this 270,896-parameter joint-group row and covers all 270,896 weight parameters; sample accuracy remains 88.2%, global posterior-to-chain-start support is 0.7389, block posterior-minus-chain remains negative at -0.0019, global posterior-minus-chain is -0.0019, and rewind remains closer by 0.0362. These rows remove the practical weight-tensor coverage objection, while literal dense CIFAR full-network covariance remains outside budget: a single float64 dense matrix would require 553.1 GiB, and matrix plus Cholesky would require 1,106.3 GiB.

4.4 Function Space Agreement Is Not Mask Equivalence

We also make the original proposal’s distributional criterion explicit. On digits, posterior sample masks fail two of the four stated thresholds: layer-sparsity KS has $p = 0.0788$ and pairwise mask-Hamming overlap is 0.631, while logit CKA passes. Mean-shift clustering in raw parameter PCA collapses posterior samples to one representative mode; the raw-parameter basin entropy is 0.0 nats with effective cluster count 1.0. The full-data CIFAR-10 ResNet-20 strengthens this separation: posterior samples can keep logit/final-hidden activation CKA above 0.93/0.91 while failing layer-sparsity, Hamming, or posterior-basin-count criteria.

Alignment and stronger direct posterior probes do not change this conclusion: activation-channel alignment are insufficient. The activation-aligned CIFAR row still fails layer KS and Hamming while retaining high activation CKA. Weight-correlation channel alignment reaches the same conclusion. The saved artifact rerun supports record-level post-hoc matching, and a block-coordinate ResNet channel audit still leaves global-channel posterior/ticket Hamming near 0.21 rather than ticket-like agreement. An exact stage-1 enumeration validates the local solver on a small ResNet subgraph, while exhaustive full-data graph isomorphism

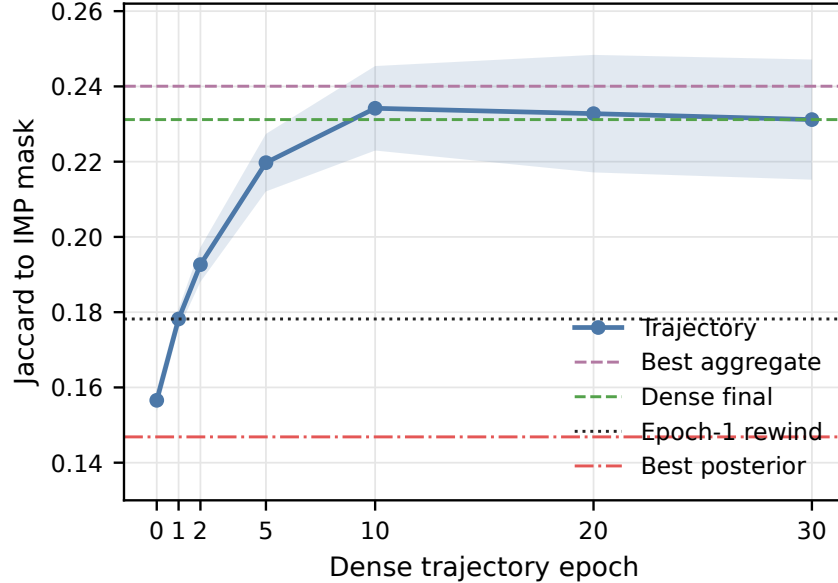


Figure 3: CIFAR-10 ResNet-20 dense-trajectory controls. Magnitude supports along the matched dense trajectory, especially around epochs 10–30, and the best aggregate trajectory magnitude score are closer to the IMP mask than any global posterior support in the movement diagnostics.

remains infeasible.

The multi-chain and high-fidelity direct rows address the concern that a single local chain was too weak. A dense-start direct probe with 75 cyclical-SGLD posterior samples fails layer KS and Hamming while maintaining high CKA; an independent-start version with 15 dense chain starts and 75 samples does the same. The rank-128 Hessian-plus-diagonal direct probe is the partial exception: 50 posterior samples now pass the Hamming-overlap threshold (0.816) and retain high logit/final-hidden activation CKA (0.932/0.910), but still fail the layer-sparsity KS threshold ($p = 2.0 \times 10^{-6}$), and all 50 samples still collapse to one parameter-PCA basin. A streamed exact joint-group direct probe over the same all 270,896 weight parameters closes the full-weight Gaussian gap: 25 joint-group posterior samples keep sample accuracy remains 88.3%, posterior-to-chain-start Hamming is 0.0503, layer-sparsity KS is $p = 1.1 \times 10^{-8}$, Hamming overlap is 0.000, logit/final-hidden activation CKA remains 0.937/0.920, and All 25 joint-group samples still collapse to one parameter-PCA basin.

4.5 IMP Residuals Are Process Objects

Posterior failure leaves a positive mechanism to explain. Dense trajectory magnitude is useful: in the matched CIFAR dense-trajectory probe, trajectory magnitude supports are closer to IMP than posterior supports, and fixed-mask training shows that final dense, epoch-10, and RMS trajectory masks are much better than random. They are still below IMP, so the remaining question is what IMP adds.

Residual swap and removal-order controls localize the gap to a process-selected IMP-only residual. Anatomy, predictor-mask, cross-seed, direct-transfer, base-compatibility, posterior-ordering, and stratified controls give the same localization. Learned mask baselines do not replace that residual: Gem-Miner-style, Bernoulli/Concrete variational pruning, and the hard-concrete L0 gate baseline are below IMP in the current CIFAR support/accuracy rows and do not rescue the calibration/OOD tradeoff. Their learned objective can change uncertainty behavior, but not into a ticket-support explanation.

The strongest process controls test whether simpler score composition explains the residual. A tensor matched excluded oracle is still below the round-selected IMP residual, and a tensor+score matched excluded oracle remains below it even after matching parameter tensor and within-tensor round-score deciles. A residualized round-score projection shows that removing base, dense-final, and final-IMP magnitude subspaces re-

duces both accuracy and oracle overlap. The original posterior-projection round-selected row reaches 88.47% accuracy. A posterior-residualized row reaches 88.25% after additionally residualizing against diagonal-Laplace posterior RMS, posterior standard deviation, and posterior RMS-minus-dense features; the gap is 0.23 percentage points, 5/5 positive seeds with 95% CI $[-0.03, 0.48]$, and overlap drops from 0.6773 to 0.4850, a 0.1923 paired drop with 5/5 positive seeds. The diagonal-Laplace posterior score subspace does not explain away the process signal. Finally, the learned trajectory/process subspace check gives the cleanest compact result: the round-selected row reaches 88.69% accuracy, the learned-subspace residualized row reaches 88.21%, the gap is 0.48 percentage points, 5/5 positive seeds with 95% CI $[0.29, 0.67]$, and oracle overlap falls from 0.6807 to 0.4917, a 0.1890 learned-subspace paired drop. Thus the current learned process subspace captures broad directions but does not substitute for the particular residual coordinates selected by IMP.

5 Discussion

These results constrain a simple Bayesian interpretation of lottery tickets. The posterior signal exists in a weak sense: posterior-induced masks are not random. But this is not enough. The same signal is already present in the dense, rewind, trajectory, or chain-start magnitude support. The explicit distribution-equivalence audit turns this into a negative result rather than a visual or single-metric impression: posterior support distributions separate from random masks, yet they are tied to or worse than local magnitude controls across the posterior families tested here. The evidence therefore favors a trajectory/magnitude subspace account plus a process-selected residual over a posterior-mode account.

This conclusion is compatible with mode-connectivity results. Independent solutions can occupy separated regions in parameter or function space, and yet the sparse supports induced by local posterior samples need not explain the IMP ticket. A generated linear connectivity audit makes this point concrete. On MNIST and Fashion-MNIST, dense-to-IMP linear barriers are nearly zero (0.0026 and 0.0395), but posterior support remains slightly below the chain-start magnitude control. On CIFAR-10 ResNet-20, the long SGLD and SWAG dense-to-IMP barriers are large (3.0827 and 3.7402), but posterior support is still tied to chain-start support. Thus linear barriers are orthogonal landscape diagnostics, not evidence of posterior-ticket equivalence.

Scope of the claim. The claim is not that Bayesian neural-network posteriors are useless, nor that no posterior approximation could ever correlate with IMP coordinates. The claim is narrower: under the posterior families and CIFAR-10/MNIST/Fashion-MNIST settings tested here, posterior-induced sparse supports do not satisfy the support-equivalence claim once dense, rewind, trajectory, and pruning-process controls are included. The positive evidence in the paper is the alternative localization: useful ticket structure is better explained by trajectory magnitude plus a process-selected IMP residual.

6 Limitations and Next Experiments

The evidence is broad, but the paper should still be read as a controlled negative result under the posterior approximations we can currently test, not as a theorem about all Bayesian neural-network posteriors. The main limitations are:

- alignment robustness. Activation-channel alignment, weight-correlation channel alignment, record-level post-hoc matching, a block-coordinate global channel audit, and exact stage-1 enumeration are negative or bounded, but exhaustive full-data graph isomorphism remains infeasible and unimplemented;
- full-network exact/full-covariance Laplace or another still broader high-fidelity posterior baseline. Current evidence includes multi-chain cyclical SGLD, rank-128 low-rank Laplace, all-weight joint-group Laplace, 20-snapshot SWAG, low-rank Hessian-plus-diagonal Laplace, tensor-block and joint-group full-covariance probes, and subspace HMC, but the exact dense full-network CIFAR posterior remains absent;

- broader learned-mask distribution variants. Gem-Miner-style score training, Bernoulli/Concrete variational pruning, and hard-concrete L0 gates do not rescue support, accuracy, calibration, or CIFAR-100 OOD behavior in the current five-seed rows, but additional permutation-aware learned-mask distributions would make this negative claim cleaner;
- additional process-causality controls. The current residual-swap, anatomy, predictor-mask, transfer, base compatibility, ordering, stratification, IMP-process, tensor+score-matched round-exclusion, residualized-score projection, posterior-residualized, and learned-subspace residualized controls do not exhaust the possible causal interventions on the pruning process;
- reproducibility hardening. The repository has a dependency list, checked environment lock, Makefile targets, artifact verifier, CI workflow, public release manifest, local release archive, source-only repository snapshot, CPU artifact-verification container, GPU training-container definition, external-validation runbook, and submission handoff. It still lacks public archive, public repository, external CI, and external GPU-run receipts.

Since the aligned, multi-chain, low-rank Laplace, and block or joint-group Laplace CIFAR runs preserve the current pattern, the next decisive empirical gap is an even stronger high-fidelity posterior baseline. If that baseline also stays near the chain-start/rewind controls, the negative result is the right paper: posterior samples reveal local magnitude structure, not IMP ticket structure. If it contradicts the current pattern, the claim should be weakened to a partial-equivalence statement that isolates the architectures, sparsities, and posterior approximations where the Bayesian account survives.

References

- [1] Samuel K. Ainsworth, Jonathan Hayase, and Siddhartha Srinivasa. Git re-basin: Merging models modulo permutation symmetries. In *International Conference on Learning Representations*, 2023.
- [2] Felix Draxler, Kambis Veschgini, Manfred Salmhofer, and Fred A. Hamprecht. Essentially no barriers in neural network energy landscape. In *International Conference on Machine Learning*, 2018.
- [3] Rahim Entezari, Hanie Sedghi, Olga Saukh, and Behnam Neyshabur. The role of permutation invariance in linear mode connectivity of neural networks. In *International Conference on Learning Representations*, 2022.
- [4] Jonathan Frankle and Michael Carbin. The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *International Conference on Learning Representations*, 2019.
- [5] Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M. Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*, 2020.
- [6] Timur Garipov, Pavel Izmailov, Dmitrii Podoprikin, Dmitry P. Vetrov, and Andrew Gordon Wilson. Loss surfaces, mode connectivity, and fast ensembling of dnns. In *Advances in Neural Information Processing Systems*, 2018.
- [7] Pavel Izmailov, Sharad Vikram, Matthew D. Hoffman, and Andrew Gordon Wilson. What are bayesian neural network posteriors really like? In *International Conference on Machine Learning*, 2021.
- [8] Nicholas Kuhn, Arvid Weyrauch, Lars Heyen, Achim Streit, Markus Götz, and Charlotte Debus. Bayesian lottery ticket hypothesis, 2026.
- [9] Namhoon Lee, Thalaiyasingam Ajanthan, and Philip H. S. Torr. SNIP: Single-shot network pruning based on connection sensitivity. In *International Conference on Learning Representations*, 2019.
- [10] Wesley J. Maddox, Pavel Izmailov, Timur Garipov, Dmitry P. Vetrov, and Andrew Gordon Wilson. A simple baseline for bayesian uncertainty in deep learning. In *Advances in Neural Information Processing Systems*, 2019.
- [11] Stephan Mandt, Matthew D. Hoffman, and David M. Blei. Stochastic gradient descent as approximate bayesian inference. *Journal of Machine Learning Research*, 2017.
- [12] Mansheej Paul, Feng Chen, Brett W. Larsen, Jonathan Frankle, Surya Ganguli, and Gintare Karolina Dziugaite. Unmasking the lottery ticket hypothesis: What’s encoded in a winning ticket’s mask? In *International Conference on Learning Representations*, 2023.
- [13] Vivek Ramanujan, Mitchell Wortsman, Aniruddha Kembhavi, Ali Farhadi, and Mohammad Rastegari. What’s hidden in a randomly weighted neural network? In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020.
- [14] Keitaro Sakamoto and Issei Sato. Analyzing lottery ticket hypothesis from pac-bayesian theory perspective. In *Advances in Neural Information Processing Systems*, 2022.
- [15] Hidenori Tanaka, Daniel Kunin, Daniel L. K. Yamins, and Surya Ganguli. Pruning neural networks without any data by iteratively conserving synaptic flow. In *Advances in Neural Information Processing Systems*, 2020.
- [16] Max Welling and Yee Whye Teh. Bayesian learning via stochastic gradient langevin dynamics. In *International Conference on Machine Learning*, 2011.
- [17] Andrew Gordon Wilson and Pavel Izmailov. Bayesian deep learning and a probabilistic perspective of generalization. In *Advances in Neural Information Processing Systems*, 2020.

Sampler	Scale	Post-chain	Post.-Chain	Rewind-Post.	Sample Acc.
DiagLap	10^{-3}	0.8826	-0.0021	0.0340	0.8799
KFACLap	10^{-3}	0.8016	-0.0016	0.0334	0.8839
KFACLap	10^{-2}	0.4859	-0.0154	0.0473	0.8695
LowRank32Lap	10^{-3}	0.9299	-0.0007	0.0345	0.8843
LowRank32Lap	10^{-2}	0.7402	-0.0098	0.0436	0.8789
LowRankLap	10^{-3}	0.9295	-0.0008	0.0330	0.8848
LowRankLap	10^{-2}	0.7359	-0.0105	0.0427	0.8784
SGHMC	10^{-7}	0.6796	-0.0037	0.0358	0.8752
SGLD	10^{-6}	0.7362	-0.0017	0.0359	0.8753
SWAG20	1.6×10^1	0.9528	-7.0e-05	0.0328	0.8636
SWAG20	6.4×10^1	0.9086	-0.0002	0.0329	0.8041
cSGLD	10^{-6}	0.7046	-0.0033	0.0368	0.8782

Table 4: Paired five-seed CIFAR movement statistics. Post.-Chain is posterior-to-IMP minus chain-start-to-IMP; Rewind-Post. is rewind-magnitude-to-IMP minus posterior-to-IMP.

Sampler	Scale	Head post-chain	Head post.-chain	Head rewind-post.	Sample Acc.
HeadLap	10^{-3}	0.7773	-0.0085	0.0209	0.8856
HeadLap	10^{-2}	0.6912	-0.0284	0.0407	0.8855
HeadLap	10^0	0.6769	-0.0352	0.0475	0.8835

Table 5: Exact full-covariance Laplace probe for the final CIFAR classification head only. Head post.-chain is head posterior-to-IMP minus head chain-start-to-IMP; head rewind-post. is head rewind-magnitude-to-IMP minus head posterior-to-IMP.

A Generated Evidence Tables

Sampler	Block	Params	B post-chain	B post.-chain	B rewind-post.	G post.-chain	Acc.
BlockDiagLap	11-block diag	22064	0.5158	-0.0114	0.0667	0.0036	0.8810
BlockDiagLap	16-block diag	68144	0.6146	-0.0050	0.0520	0.0010	0.8802
BlockLap	layer1.0.conv1	2304	0.2236	-0.0075	0.0464	0.0013	0.8961
BlockLap	layer3.0.shortcut.0	2048	0.3626	-0.0010	0.0648	0.0008	0.8905
JointDiagLap	joint-group diag 20k	86576	0.7009	-0.0023	0.0465	0.0006	0.8828
JointDiagLap	joint-group diag 10k	68144	0.5804	-0.0050	0.0523	0.0015	0.8811
JointDiagLap	joint-group diag 40k	270896	0.7389	-0.0019	0.0362	-0.0019	0.8824
JointLap	joint group	5424	0.5088	-0.0206	0.0343	0.0010	0.8922

Table 6: Full-covariance block Laplace probes for selected CIFAR ResNet-20 weight tensors, tensor groups, or a tensor-block diagonal subset with the rest of the network frozen. Block post.-chain is block posterior-to-IMP minus block chain-start-to-IMP; global post.-chain reports the induced global support change.

Sampler	Step	PostCh.	Post.-Chain	Rwd.-Post.	Acc.	Accept	ParamDist.
Hess16SubHMC	3×10^{-4}	0.9994	-1.6e-05	0.0331	0.8881	0.8833	0.0095
Hess32SubHMC	3×10^{-4}	0.9993	3.6e-05	0.0326	0.8872	0.9500	0.0104
HessSubHMC	3×10^{-4}	0.9999	2.3e-06	0.0339	0.8865	0.8600	0.0022
RandSubHMC	3×10^{-3}	0.9766	-2.2e-05	0.0339	0.8863	0.7400	0.3672
TrajSubHMC	3×10^{-4}	0.9950	-0.0001	-0.0499	0.8849	0.8000	0.0803
TrajSubHMC	10^{-3}	0.9915	-0.0002	-0.0498	0.8847	0.6900	0.1589

Table 7: Full-network low-dimensional subspace HMC probes around the dense CIFAR checkpoint. RandSubHMC and TrajSubHMC use random and dense-trajectory subspaces; Hessian rows use randomized top-Hessian subspaces.

Row	Post.	Chain	Delta	Wins	Post-chain	W-dist.
JointLap	0.3294	0.3501	-0.0206	0.0000	0.5088	0.0206
DiagLap	0.1447	0.1469	-0.0021	0.0000	0.8826	0.0021
HeadLap	0.6983	0.7068	-0.0085	0.4000	0.7773	0.0125
Hess16HMC	0.1468	0.1468	-1.6e-05	0.4000	0.9994	3.8e-05
Hess32HMC	0.1461	0.1461	3.6e-05	0.8000	0.9993	3.7e-05
KFACLap	0.1456	0.1456	-2.2e-05	0.6000	0.9334	0.0002
LowRank32Lap	0.1358	0.1457	-0.0098	0.0000	0.7402	0.0098
LowRankLap	0.1351	0.1456	-0.0105	0.0000	0.7359	0.0105
RandHMC	0.1440	0.1440	-2.2e-05	0.2000	0.9766	6.8e-05
SGHMC move	0.1419	0.1457	-0.0037	0.0000	0.6796	0.0037
SGLD move	0.1425	0.1441	-0.0017	0.0000	0.7362	0.0017
SGLD-3	0.1368	0.1368	-4.9e-06	0.4000	0.9969	3.4e-05
SWAG	0.1361	0.1361	-5.4e-05	0.6000	0.9265	0.0003
SWAG20	0.1454	0.1455	-7.0e-05	0.2000	0.9528	0.0001
TrajHMC	0.2290	0.2292	-0.0002	0.4000	0.9915	0.0002
cSGLD move	0.1422	0.1454	-0.0033	0.0000	0.7046	0.0033
Fashion	0.2114	0.2122	-0.0008	0.0000	0.9212	0.0008
MNIST	0.2373	0.2385	-0.0011	0.0000	0.9298	0.0011

Table 8: Mode/ticket distribution-equivalence audit for selected posterior families. Delta is posterior-to-IMP minus matched chain-start-to-IMP. Wins is the fraction of seed-level paired posterior-chain deltas above zero; W-dist. is the two-sample Wasserstein distance over grouped seed means. A positive posterior mode account would require positive deltas rather than practical ties to chain-start magnitude supports.

Setting	Comparison	Modes	H_n	n_L	n_T	KS p	Ham.	Logit	Act. H-cost	A-cost	Pass
Digits MLP	posterior samples vs tickets	1	0.0000	50	5	0.0788	0.6314	0.9819	n/a	0.0181	n/a 2/4
Digits MLP	posterior modes vs tickets	1	0.0000	1	5	0.9216	n/a	0.9821	n/a	0.0179	n/a 3/4
CIFAR subset ResNet	posterior samples vs tickets	1	0.0000	25	5	1.1e-05	0.6000	0.8878	n/a	0.1122	n/a 2/4
CIFAR subset ResNet	posterior modes vs tickets	1	0.0000	1	5	0.1388	n/a	0.9077	n/a	0.0923	n/a 3/4
CIFAR subset act. pilot	posterior samples vs tickets	3	0.4057	15	3	0.0330	0.6571	0.9022	0.8726	0.0978	0.1274 4/6
CIFAR subset act. pilot	posterior modes vs tickets	3	0.4057	3	3	0.2264	1.0000	0.9014	0.8704	0.0986	0.1296 6/6
CIFAR full ResNet	posterior samples vs tickets	1	0.0000	50	5	5.3e-09	0.0033	0.9369	0.9172	0.0631	0.0828 4/6
CIFAR full ResNet	posterior modes vs tickets	1	0.0000	1	5	0.0329	n/a	0.9383	0.9179	0.0617	0.0821 4/6
CIFAR full aligned	aligned posterior samples vs tickets	1	0.0000	50	5	2.3e-09	0.0000	0.9373	0.9168	0.0627	0.0832 4/6
CIFAR full aligned	aligned posterior modes vs tickets	1	0.0000	1	5	0.0413	n/a	0.9363	0.9160	0.0637	0.0840 4/6
CIFAR full weight-aligned	weight-aligned chain start magnitude vs tickets	1	0.0000	5	5	4.8e-05	0.0000	0.9374	0.9195	0.0626	0.0805 4/6
CIFAR full weight-aligned	weight-aligned posterior samples vs tickets	1	0.0000	50	5	1.2e-08	0.1290	0.9336	0.9131	0.0664	0.0869 4/6
CIFAR full weight-aligned	weight-aligned posterior modes vs tickets	1	0.0000	1	5	0.0413	n/a	0.9327	0.9132	0.0673	0.0868 4/6
CIFAR full cSGLD multi-chain	chain start magnitude vs tickets	1	0.0000	15	5	5.2e-07	0.0857	0.9363	0.9196	0.0637	0.0804 4/6
CIFAR full cSGLD multi-chain	posterior samples vs tickets	1	0.0000	75	5	3.3e-08	0.2461	0.9327	0.9144	0.0673	0.0856 4/6
CIFAR full cSGLD multi-chain	posterior modes vs tickets	1	0.0000	1	5	0.0413	n/a	0.9288	0.9115	0.0712	0.0885 4/6
CIFAR full cSGLD independent	chain start magnitude vs tickets	1	0.0000	15	5	7.6e-08	0.0000	0.9306	0.9105	0.0694	0.0895 4/6
CIFAR full cSGLD independent	posterior samples vs tickets	1	0.0000	75	5	9.3e-10	0.0000	0.9269	0.9051	0.0731	0.0949 4/6
CIFAR full cSGLD independent	posterior modes vs tickets	1	0.0000	1	5	0.0413	n/a	0.9242	0.9054	0.0758	0.0946 4/6
CIFAR full LowRank128Lap	chain start magnitude vs tickets	1	0.0000	5	5	9.0e-05	0.0000	0.9385	0.9218	0.0615	0.0782 4/6
CIFAR full LowRank128Lap	posterior samples vs tickets	1	0.0000	50	5	2.0e-06	0.8163	0.9319	0.9096	0.0681	0.0904 5/6
CIFAR full LowRank128Lap	posterior modes vs tickets	1	0.0000	1	5	0.1388	n/a	0.9289	0.9023	0.0711	0.0977 5/6
CIFAR full JointDiagLap270k	chain start magnitude vs tickets	1	0.0000	5	5	2.5e-05	0.0000	0.9389	0.9221	0.0611	0.0779 4/6
CIFAR full JointDiagLap270k	posterior samples vs tickets	1	0.0000	25	5	1.1e-08	0.0000	0.9373	0.9199	0.0627	0.0801 4/6
CIFAR full JointDiagLap270k	posterior modes vs tickets	1	0.0000	1	5	0.0413	n/a	0.9379	0.9225	0.0621	0.0775 4/6

Table 9: Direct proposal-level mode/ticket distribution probe on digits MLP plus CIFAR-10 ResNet-20 subset and full-data settings. Posterior sample masks and mean-shift posterior-mode representatives are compared with IMP tickets using the proposal’s layer-sparsity KS, mask-Hamming overlap, logit-space CKA, final-hidden-activation CKA, and Hungarian-cost thresholds; Modes and H_n report mean-shift cluster count and normalized raw-parameter basin entropy.

Source	Acc.	NLL	ECE	MSP AUROC	Ent. AUROC	MSP FPR95
Dense	0.8866	0.3536	0.0353		0.8230	0.8326
IMP	0.8953	0.3387	0.0393		0.8306	0.8392
SWAG ens.	0.8688	0.4018	0.0285		0.8050	0.8171
Learned random	0.8449	0.4632	0.0270		0.7897	0.8024
Gem-Miner	0.8418	0.4694	0.0283		0.7853	0.7985
Var. prune	0.8301	0.5008	0.0255		0.7754	0.7897

Table 10: Five-seed CIFAR-10 calibration and CIFAR-100 OOD diagnostics. SWAG ens. averages ten SWAG predictive samples per seed. Learned random, Gem-Miner, and Var. prune rows are trained from initialization at the IMP sparsity. MSP and entropy AUROC use maximum softmax probability and negative predictive entropy as in-distribution scores; lower FPR95 is better.

Source	Acc.	Support-IMP	Support-Dense	Support-Rewind
Epoch 1	0.4017		0.2231	1.0000
Epoch 5	0.6881	0.1782	0.3582	0.3724
Epoch 10	0.7630	0.2342	0.5350	0.2746
Epoch 20	0.8492	0.2328	0.8433	0.2273
Epoch 30	0.8841	0.2312	1.0000	0.2231
max abs	—	0.2342	0.5616	0.3517
mean abs	—	0.2388	0.5314	0.4214
rms abs	—	0.2400	0.5713	0.4046

Table 11: Five-seed CIFAR dense-trajectory controls. The epoch-1 state is the IMP rewind point. Checkpoint rows use magnitude supports at that epoch; aggregate rows use trajectory score masks across checkpoints.

Source	Kind	Accuracy	Acc.-IMP	Support-IMP
IMP	IMP	0.8983	0.0006	1.0000
random 0	random	0.8422	-0.0555	0.0922
Epoch 1	checkpoint	0.8540	-0.0437	0.1785
Epoch 10	checkpoint	0.8730	-0.0247	0.2374
Epoch 30	checkpoint	0.8826	-0.0151	0.2356
rms abs	aggregate	0.8739	-0.0238	0.2429
mean abs	aggregate	0.8743	-0.0234	0.2421
path length	aggregate	0.8549	-0.0428	0.1772
rewind rms movement	aggregate	0.8572	-0.0405	0.1639
Gem-Miner	Gem-Miner	0.8471	-0.0499	0.0917
Var. prune	Var. prune	0.8306	-0.0669	0.0907
Hard concrete	Hard concrete	0.2766	-0.6204	0.0922

Table 12: Five-seed CIFAR trajectory-mask retraining controls. All masks are trained for 30 epochs from the same epoch-1 rewind state at the IMP sparsity. Acc.-IMP is candidate accuracy minus the final IMP accuracy from the corresponding seed.

Source	Kind	Acc.	ECE	Brier	Acc.-IMP	Supp.-IMP	Supp.-Dense
IMP	IMP	0.9711	0.0211	0.0410	0.0000	1.0000	0.7604
random 0	random	0.9489	0.0398	0.0821	-0.0222	0.2151	0.2139
Gem-Miner	Gem-Miner	0.9550	0.0392	0.0753	-0.0161	0.2081	0.2082
Var. prune	Var. prune	0.9633	0.0250	0.0532	-0.0078	0.3822	0.4010

Table 13: Five-seed digits MLP variational pruning sanity check. Var. prune optimizes Bernoulli mask probabilities on frozen initialization weights with Concrete samples, KL, sparsity, and entropy penalties before retraining the selected hard mask.

Base	Variant	Alpha	Accuracy	Acc.-IMP
Epoch 30	top IMP	0	0.8797	-0.0175
Epoch 30	top IMP	0.5	0.8882	-0.0090
Epoch 30	random non-IMP	0.5	0.8780	-0.0192
Epoch 30	top IMP	1	0.8966	-0.0006
rms abs	top IMP	0	0.8733	-0.0238
rms abs	top IMP	0.5	0.8851	-0.0121
rms abs	random non-IMP	0.5	0.8705	-0.0266
Epoch 10	top IMP	0	0.8712	-0.0259
Epoch 10	top IMP	0.5	0.8855	-0.0116
Epoch 10	random non-IMP	0.5	0.8704	-0.0268

Table 14: Five-seed CIFAR trajectory residual controls. Starting from a base trajectory mask, an IMP-residual row swaps the indicated fraction of base-only support for IMP-only support; a random-residual row swaps the same number of positions for non-IMP random support.

Base	Base	Low rm	Rand. rm	High rm	Rand. non
Epoch 30	0.8792	0.8881	0.8883	0.8906	0.8779
rms abs	0.8738	0.8874	0.8896	0.8914	0.8709
Epoch 10	0.8687	0.8862	0.8922	0.8920	0.8701

Table 15: Five-seed CIFAR residual removal-order controls. Low/Rand./High rm add the same top IMP-only residual support and vary which base-only weights are removed by low, random, or high base score. Rand. non removes low base-score weights but adds same-size non-IMP random support.

Base	Support-IMP	IMP-only(k)	Prune rnd.	Pred. AUC	Pred. recall
Epoch 30	0.2323	28.4k	2.9003	0.6197	0.2175
rms abs	0.2412	27.8k	2.9745	0.6165	0.2087
Epoch 10	0.2359	28.1k	2.9669	0.6206	0.2206

Table 16: Five-seed CIFAR residual anatomy. IMP-only(k) is the number of IMP-kept weights missing from the base support. Prune rnd. is the mean IMP pruning round for base-only weights. The predictor is a held-out logistic model over dense-trajectory rank features plus stage indicators, trained to recover IMP-only residual support among non-base weights.

Base	Base	Oracle	Pred.	Random	Pred. prec.	Rand. prec.
Epoch 30	0.8808	0.8892	0.8793	0.8805	0.1866	0.1253
rms abs	0.8740	0.8866	0.8744	0.8744	0.1834	0.1233
Epoch 10	0.8712	0.8890	0.8728	0.8724	0.1843	0.1239

Table 17: Five-seed CIFAR functional residual-predictor masks. For each base, the held-out predictor swaps half of base-only support for non-base weights with highest predicted IMP-only probability. Oracle uses true IMP-only residual support, and random uses same-size held-out non-IMP-random support. Prec. is the IMP-only precision of added weights.

Base	Base	Oracle	Cross-seed	Random	Cross prec.	Rand. prec.
Epoch 30	0.8815	0.8905	0.8776	0.8781	0.2413	0.1260
rms abs	0.8733	0.8878	0.8731	0.8725	0.2238	0.1246
Epoch 10	0.8730	0.8890	0.8745	0.8726	0.2388	0.1264

Table 18: Five-seed CIFAR cross-seed residual-transfer masks. For each held-out target seed, the predictor is trained on non-base candidate weights from the other four seeds, then swaps half of target base-only support for target non-base weights with highest predicted IMP-only probability. Oracle and random use the target seed only.

Base	Base	Oracle	Src vote	Aln src	Vote rand.	Aln rand.	Tgt rand.	Src prec.
Epoch 30	0.8813	0.8886	0.8769	0.8797	0.8774	0.8809	0.8779	0.1513
rms abs	0.8737	0.8872	0.8739	0.8727	0.8735	0.8743	0.8728	0.1451
Epoch 10	0.8720	0.8890	0.8725	0.8710	0.8732	0.8740	0.8719	0.1491

Table 19: Five-seed CIFAR direct cross-seed residual-support transfer. Source-vote variants add target non-base coordinates selected by votes from the other seeds’ oracle residual supports. Aln src and Aln rand. first map source channels to target channels by activation-correlation Hungarian matching. Tgt rand. is a target-seed random non-base residual control. Src prec. is target IMP-only precision among source-vote additions.

Base	T base	T oracle	T rand.	M base	M oracle	M rand.
Epoch 30	0.8827	0.8893	0.8791	0.8641	0.8926	0.8649
rms abs	0.8743	0.8892	0.8745	0.8605	0.8910	0.8628
Epoch 10	0.8744	0.8892	0.8712	0.8607	0.8942	0.8636

Table 20: Five-seed CIFAR residual base-compatibility controls. T rows use the actual trajectory base. M rows replace that base with a per-parameter random base preserving the same IMP/non-IMP counts and therefore the same base-to-IMP overlap before adding oracle or random residual support.

Base	M base	Top	Dense	RMS	RMS-d	Std	Rand	Dense ovlp	RMS ovlp	Rand ovlp
Epoch 30	0.8606	0.8915	0.8821	0.8852	0.8770	0.8710	0.8783	0.5573	0.5560	0.5006
rms abs	0.8631	0.8928	0.8812	0.8834	0.8745	0.8717	0.8795	0.5528	0.5510	0.5011
Epoch 10	0.8609	0.8911	0.8827	0.8829	0.8764	0.8710	0.8791	0.5565	0.5556	0.5002

Table 21: Five-seed CIFAR residual posterior-decomposition controls on IMP-overlap-matched random bases. Top adds the target oracle IMP-only residual. Dense, RMS, RMS-d, Std, and Rand add the same number of target IMP-only residual weights ranked by dense final magnitude, diagonal-Laplace posterior RMS, posterior RMS minus dense magnitude, posterior standard deviation, or uniform sampling. Ovp columns report the fraction of additions that also belong to the top oracle additions.

Base	Base	Oracle	Rand.	IMP	Global non	Param+score non
Epoch 30	0.8803	0.8872		0.8818	0.8767	0.8758
rms abs	0.8708	0.8858		0.8794	0.8701	0.8716
Epoch 10	0.8678	0.8854		0.8764	0.8695	0.8685

Table 22: Five-seed CIFAR residual stratified controls. All non-base variants remove the same low-base-score weights as the oracle. Rand. IMP adds random IMP-only residual weights. Global non adds random non-IMP non-base weights. Param+score non matches the oracle added weights by parameter tensor and within-parameter score decile, but excludes IMP-only weights.

Base	Base	Oracle	R1 acc.	R3 acc.	R5 acc.	R1 prec.	R3 prec.
Epoch 30	0.8793	0.8898	0.8792	0.8829	0.8867	0.4431	0.7523
rms abs	0.8736	0.8881	0.8771	0.8832	0.8857	0.4308	0.7459
Epoch 10	0.8724	0.8884	0.8772	0.8826	0.8861	0.4395	0.7536

Table 23: Five-seed CIFAR residual IMP-process probe. R1/R3/R5 rows add half residual support from IMP round-survivor candidates ranked by the corresponding round-trained weights. Prec. is the fraction of added weights that survive in the final IMP mask; R5 precision is one by construction.

Base	Rnd	Top acc.	Rand. acc.	Low acc.	Top final	Rand. final	Low final	Top oracle	Rand. oracle	Low oracle
Epoch 30	1	0.8806	0.8774	0.8776	0.4373	0.1913	0.0811	0.2706	0.0960	0.0340
Epoch 30	3	0.8839	0.8813	0.8770	0.7530	0.4357	0.1881	0.4777	0.2176	0.0652
Epoch 30	5	0.8887	0.8847	0.8815	1.0000	1.0000	1.0000	0.6818	0.4998	0.3182
rms abs	1	0.8788	0.8733	0.8738	0.4268	0.1871	0.0808	0.2617	0.0924	0.0343
rms abs	3	0.8830	0.8759	0.8726	0.7496	0.4294	0.1885	0.4734	0.2152	0.0657
rms abs	5	0.8857	0.8803	0.8759	1.0000	1.0000	1.0000	0.6803	0.4987	0.3197
Epoch 10	1	0.8795	0.8724	0.8699	0.4358	0.1880	0.0818	0.2698	0.0933	0.0345
Epoch 10	3	0.8822	0.8746	0.8727	0.7555	0.4347	0.1898	0.4795	0.2178	0.0657
Epoch 10	5	0.8855	0.8805	0.8755	1.0000	1.0000	1.0000	0.6821	0.4992	0.3179

Table 24: Five-seed CIFAR residual IMP-process ranking controls. Top uses the highest round-trained scores among round-survivor candidates; Rand. samples the same round-survivor candidate set; Low uses the lowest round-trained scores. Final is the fraction of added weights that survive in the final IMP mask; oracle is overlap with the final oracle added subset.

Base	Rnd	Score acc.	Match acc.	Δ	Wins	Oracle
Epoch 10	1	0.8823	0.8799	0.0024	5/5	0.5568
Epoch 10	3	0.8825	0.8789	0.0036	4/5	0.5976
Epoch 10	5	0.8851	0.8820	0.0030	5/5	0.6792
Epoch 30	1	0.8841	0.8843	-0.0002	3/5	0.5569
Epoch 30	3	0.8842	0.8831	0.0011	4/5	0.5971
Epoch 30	5	0.8860	0.8845	0.0014	3/5	0.6788
rms abs	1	0.8824	0.8787	0.0038	4/5	0.5538
rms abs	3	0.8821	0.8808	0.0013	3/5	0.5943
rms abs	5	0.8854	0.8836	0.0018	4/5	0.6773

Table 25: Five-seed CIFAR residual IMP-process oracle-overlap-matched control. Score acc. uses the round-trained score ordering inside the final-IMP residual candidate set. Match acc. samples random final-IMP residual additions with the same final-oracle overlap count. Δ is score minus matched random accuracy; wins are positive paired seed deltas.

Base	Rnd	Round acc.	Dense acc.	Base acc.	Δ_D	WinD	Δ_B	WinB	Oracle
Epoch 10	1	0.8833	0.8827	0.8824	0.0005	3/5	0.0008	4/5	0.5559
Epoch 10	3	0.8838	0.8827	0.8824	0.0011	4/5	0.0014	4/5	0.5984
Epoch 10	5	0.8872	0.8827	0.8824	0.0045	5/5	0.0048	5/5	0.6780
Epoch 30	1	0.8861	0.8831	0.8831	0.0030	5/5	0.0030	5/5	0.5574
Epoch 30	3	0.8874	0.8831	0.8831	0.0042	4/5	0.0042	4/5	0.5992
Epoch 30	5	0.8873	0.8831	0.8831	0.0041	5/5	0.0041	5/5	0.6779
rms abs	1	0.8820	0.8823	0.8820	-0.0004	2/5	-4.0e-05	2/5	0.5536
rms abs	3	0.8844	0.8823	0.8820	0.0021	4/5	0.0024	5/5	0.5950
rms abs	5	0.8870	0.8823	0.8820	0.0046	5/5	0.0050	5/5	0.6760

Table 26: Five-seed CIFAR residual IMP-process score-source controls. Candidate set is final-IMP residual for all variants; dense/base controls rank that candidate set by dense-final or base-source magnitudes. Δ_D/Δ_B are round-score minus dense/base score accuracy; wins are positive paired seed deltas.

Base	Rnd	Round acc.	Excl. acc.	Δ	Wins	Round oracle	Excl. oracle
Epoch 10	1	0.8844	0.8770	0.0074	5/5	0.5602	0.4398
Epoch 10	3	0.8858	0.8782	0.0076	5/5	0.5995	0.4005
Epoch 10	5	0.8872	0.8779	0.0093	5/5	0.6762	0.3238
Epoch 30	1	0.8848	0.8830	0.0018	4/5	0.5608	0.4392
Epoch 30	3	0.8867	0.8834	0.0033	5/5	0.5983	0.4017
Epoch 30	5	0.8867	0.8824	0.0043	5/5	0.6755	0.3245
rms abs	1	0.8846	0.8779	0.0067	5/5	0.5581	0.4419
rms abs	3	0.8858	0.8777	0.0081	5/5	0.5959	0.4041
rms abs	5	0.8861	0.8793	0.0068	5/5	0.6733	0.3267

Table 27: Five-seed CIFAR residual IMP-process round-exclusion control. Round acc. uses the round-trained score ordering inside the final-IMP residual candidate set. Excl. acc. removes the round-selected additions from that candidate set, then chooses the best remaining final-IMP residual additions by final IMP magnitude under the same support budget. Δ is round minus excluded accuracy; wins are positive paired seed deltas.

Base	Rnd	Round acc.	Excl. acc.	Tensor excl.	Δ_E	WinE	Δ_T	WinT	Tensor oracle
rms abs	5	0.8855	0.8753	0.8764	0.0103	5/5	0.0091	5/5	0.4153

Table 28: Five-seed CIFAR residual IMP-process tensor-matched round-exclusion control at the RMS trajectory base and round 5. Tensor excl. removes the round-selected additions, then chooses replacement final-IMP residual additions matched to the removed coordinates by parameter tensor before applying the final IMP magnitude score. Δ_E is round minus the unrestricted excluded oracle; Δ_T is round minus the tensor-matched excluded oracle.

Base	Rnd	Round acc.	Tensor excl.	Tensor+score excl.	Δ_T	WinT	Δ_{TS}	WinTS
rms abs	5	0.8878	0.8784	0.8837	0.0095	5/5	0.0041	5/5

Table 29: Five-seed CIFAR residual IMP-process tensor+score-matched round-exclusion control at the RMS trajectory base and round 5. Tensor+score excl. removes the round-selected additions, then chooses replacement final-IMP residual additions matched to the removed coordinates by parameter tensor and within-tensor round-score decile before applying final IMP magnitude. Δ_T is round minus the tensor-matched excluded oracle; Δ_{TS} is round minus the tensor+score-matched excluded oracle.

Base	Rnd	Round acc.	Resid. acc.	Dense acc.	Base acc.	Δ_R	WinR	Oracle drop
rms abs	5	0.8852	0.8811	0.8814	0.8811	0.0041	5/5	0.1830

Table 30: Five-seed CIFAR residual IMP-process projection control at the RMS trajectory base and round 5. Resid. acc. uses a round-score ordering after linearly projecting out the base-source, dense-final, and final-IMP magnitude scores inside the final-IMP residual candidate pool. Δ_R is round minus residualized-score accuracy; Oracle drop is round oracle overlap minus residualized oracle overlap.

Base	Rnd	Round acc.	Post-resid. acc.	Dense acc.	Base acc.	Δ_P	WinP	Oracle drop
rms abs	5	0.8847	0.8825	0.8815	0.8797	0.0023	5/5	0.1923

Table 31: Five-seed CIFAR residual IMP-process posterior-projection control at the RMS trajectory base and round 5. Post-resid. acc. uses a round-score ordering after linearly projecting out the base-source, dense-final, final-IMP magnitude, and diagonal-Laplace posterior RMS/std/excess scores inside the final-IMP residual candidate pool. Δ_P is round minus posterior-residualized accuracy; Oracle drop is round oracle overlap minus posterior-residualized oracle overlap.

Base	Rnd	Round acc.	Subspace-resid. acc.	Dense acc.	Base acc.	Δ_S	WinS	Oracle drop
rms abs	5	0.8869	0.8821	0.8822	0.8822	0.0048	5/5	0.1890

Table 32: Five-seed CIFAR residual IMP-process learned-subspace control at the RMS trajectory base and round 5. Subspace-resid. acc. uses a round-score ordering after projecting out the top rank-8 PCA component scores learned from dense trajectory, final-IMP magnitude, and earlier IMP-round scores inside the final-IMP residual candidate pool. Δ_S is round minus learned-subspace-residualized accuracy; Oracle drop is round oracle overlap minus learned-subspace-residualized oracle overlap.